

Covert MISO-VLC Systems: Leveraging Optical Reconfigurable Intelligent Surfaces for Enhanced Security

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Abstract—As an advanced privacy-preserving solution for indoor wireless networks, covert visible light communication (VLC) can safeguard confidential data by obscuring transmission traces from surveillance entities. This paper investigates a covert multiple-input single-output (MISO)-VLC system enhanced by dual optical reconfigurable intelligent surfaces (RISs). The system comprises multiple access points (APs), a legitimate user (Bob), and a warden (Willie), where both the mirror-based RIS (mRIS) and liquid crystal-based RIS (LC-RIS) are deployed to reconfigure the optical propagation environment to improve covertness. Based on binary hypothesis testing and information-theoretic principles, we derive closed-form covertness constraints for two distinct scenarios: (i) considering peak power constraints only, and (ii) incorporating both peak and average power constraints. Subject to these constraints, we formulate an achievable rate maximization problem that jointly optimizes orientation configuration of the mRIS, the refractive index of the LC-RIS, and the AP-mRIS association. By exploiting optical transmission characteristics, the problem is decoupled into three tractable subproblems and solved via a low-complexity Geometrically-Driven Decoupling Optimization (GDDO) algorithm. Simulation results demonstrate that the proposed dual-RIS-assisted scheme achieves 10 times higher covert rates than single-RIS and non-RIS baselines.

Index Terms—Visible light communication, covert communication, reconfigurable intelligent surfaces, mirror array.

I. INTRODUCTION

VISIBLE Light Communication (VLC), as an emerging wireless technology, utilizes the visible light spectrum (380-780 nm) from Light-Emitting Diodes (LEDs) for data transmission, enabling the dual purposes of illumination and communication. With its high bandwidth density, Electromagnetic Interference (EMI) immunity, and zero occupation of scarce Radio Frequency (RF) resources, VLC has emerged as a critical technology for high-security-demand scenarios

such as smart homes, industrial IoT, and smart healthcare [2]. However, due to the inherent divergence of VLC, confidential information is highly susceptible to eavesdropping or hijacking.

To address the security challenges in VLC, some existing works primarily employ Physical-Layer Security (PLS) techniques that leverage encoding or signal processing to provide security protection [3], [4], [5], [6], [7]. For example, considering non-negativity and average optical intensity constraints, lower and upper bounds on the secrecy capacity were derived for VLC systems in [4]. To mitigate the wiretap effect of both active and passive eavesdroppers, zero-forcing beamforming strategies were proposed in [7]. Besides, the authors of [6] proposed a novel VLC PLS scheme by combining uplink positioning and Multiple-Input Multiple-Output (MIMO) beamforming to collaboratively enhance data security and location-sensing capabilities. However, these methods fail to conceal the existence of communication. Thus, malicious eavesdroppers may still launch attacks by detecting the presence of signals.

To address this, covert communication technology manipulates the statistical characteristics of signals, preventing eavesdroppers from discriminating between communication signals and background noise [8]. Avoiding detection of communication activities, covert communication is regarded as a novel direction in addressing VLC security challenges. A theoretical framework was proposed to show the fundamental performance limit for an indoor covert VLC system in [9]. For an Unmanned Aerial Vehicle (UAV)-based covert VLC system, the height and peak optical intensity were optimized in [10] and [11]. For a water-air optical communication system, a covert transmission scheme was proposed and a covert region was established to restrict the warden's detection range [12]. Besides, a gammatone filter bank-based feature extraction method was proposed to hide information adaptively for VLC systems in [13]. In summary, the research on covert VLC has achieved notable advances through establishing fundamental theoretical limits [9], exploring applications in diverse platforms like UAVs [10], [11] and specialized channels [12], and developing novel physical-layer covert signaling techniques [13].

Reconfigurable Intelligent Surfaces (RISs), a promising solution for Sixth-Generation (6G) networks, have attracted extensive research attention. By reconfiguring the wireless propagation environment, RIS is expected to enhance the

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covertness of wireless communication through improved channel manipulation. In the field of RF communications, comprehensive research has demonstrated that optimizing RIS correlation coefficients can significantly enhance covertness [14], [15], [16], [17], [18], [19]. For example, to intelligently adjust the reflecting coefficients of RIS elements, passive beamforming optimization was proposed for RF covert communications in [17]. The authors of [18] proposed two RIS-assisted finite-blocklength covert communication schemes, employing equal and unequal transmission prior probabilities respectively to maximize effective covert throughput. Besides, subject to covert requirements, the covert transmission rate is maximized by alternatively optimizing the transmitting power, the RIS phase shift, and the horizontal location of the UAV-mounted RIS in [19].

Due to fundamental differences in propagation characteristics between RF and VLC, existing research on RIS in the RF systems cannot be directly extended to VLC applications. From the hardware perspective, there are three types of optical RIS in VLC systems [20]. The first type is the mirror array-based RIS (mRIS), where each mirror unit can rotate bidirectionally to intelligently manipulate reflections of VLC links [21]. The second type is the Liquid Crystal-based RIS (LC-RIS), which controls light steering and amplification by regulating its refractive index through external electric field control [22]. The third type is the programmable metasurface-based RIS, which modifies the optical wireless environment by adjusting the phase gradient of each metasurface element [23]. Considering the relatively high cost of metasurfaces, mRIS and LC-RIS have shown promising application prospects in existing studies. For example, for an mRIS-aided Multiple-Input Single-Output (MISO)-VLC system, asymptotic capacity was investigated in [24]. For an LC-RIS-assisted VLC receiver, transmission rate was investigated and maximized in [25]. The authors of [26] and [27] deployed an mRIS within the indoor environment and integrated an LC-RIS into the receiver to mitigate the LoS blockage and enhance the data rate. Regarding RIS-assisted security enhancement in VLC systems, Ref. [28] enhances PLS in VLC systems by optimizing mRIS-element orientations to maximize achievable secrecy rate while maintaining illumination uniformity. Furthermore, our prior work [29] advances this domain by proposing an optimization framework for covert Single-Input Single-Output (SISO)-VLC systems through joint adjustments of mRIS rotation configurations and LC-RIS setups.

However, transitioning from the well-studied covert SISO-VLC systems [9], [12], [29] to practical MISO scenarios is non-trivial and introduces fundamental mathematical challenges that render existing SISO-based analyses inapplicable. Specifically, the derivation of covertness constraints becomes analytically intractable in MISO architectures where the warden receives superimposed signals from multiple Access Points (APs). Under the inherent non-negativity and amplitude constraints of VLC, the statistical distribution of such composite signals exhibits extreme complexity, making the derivation

of closed-form detection error probabilities an open problem that has not been addressed in prior literature.

Motivated by the advancements in RIS-enhanced covert communications in the RF domain and the practical importance of multi-AP scenarios, this paper investigates a dual-RIS-assisted covert VLC system composed of multiple APs, a legitimate user, and a warden. The main contributions of this work are summarized below.

- We propose a novel covert communication framework for MISO-VLC systems that integrates mRIS in the propagation channel and LC-RIS at the receiver. By establishing a hybrid channel model incorporating both geometric optical reflection and liquid crystal refraction, we leverage the joint gains of dual RISs to enhance the legitimate signal.
- We derive closed-form covertness constraints for MISO-VLC systems for the first time. By establishing tight upper bounds on the Kullback-Leibler (KL) divergence, a fundamental metric for quantifying covertness, we circumvent the analytical intractability of the superimposed signal's Probability Density Function (PDF) at the warden.
- To quantify the impact of power constraints on covert performance, we provide covertness constraints for two distinct and practical power regimes: (i) a peak-power-limited scenario, where the optimal input follows a uniform distribution, and (ii) a joint peak-and-average-power-limited scenario, governed by a more complex truncated exponential distribution.
- To maximize the covertness enhancement brought by optical RISs, we develop a comprehensive hybrid channel model for a dual-RIS-assisted MISO-VLC system and formulate achievable covert rate maximization problems incorporating: (i) different power constraints, (ii) LC-RIS refractive indices, (iii) mRIS rotation configurations, and (iv) AP-mRIS association.
- Considering the formulated achievable covert rate maximization problems are non-convex large-scale Mixed-Integer Non-Linear Programming (MINLP) problems, we propose a low-complexity Geometrically-Driven Decoupling Optimization (GDDO) algorithm to solve them. By stepwise decoupling the optimization variables, we solve AP-mRIS association by leveraging the principle of AP proximity selection and obtain the optimal mRIS configuration by the reflected spot searching.
- In the numerical examples, we demonstrate that the proposed dual-RIS-assisted VLC system achieves up to a tenfold increase in covert rate compared to single-RIS and non-RIS baselines. In addition, the simulation results also characterize the fundamental trade-offs among the covert rate, the number of mRIS elements, and the warden's position, providing quantitative guidelines for the practical deployment of covert VLC systems.

The remainder of this article is organized as follows. Section II develops a comprehensive hybrid channel model for the dual-RIS-assisted MISO-VLC system, including the LoS channel, RIS-reflected channel, and channel gain of the LC-RIS-aided receiver. Section III derives the fundamental

covert constraint for this system under two power constraints: peak power constraint and joint peak-and-average power constraint. Section IV proposes the GDDO algorithm to solve the MINLP problem for efficient covert transmission rate maximization. Numerical results evaluating the proposed algorithm are presented in Section V, and conclusions are provided in Section VI.

Notations: In this article, normal letters (e.g., A or ξ) denote scalars. Boldface letters (e.g., $\mathbf{Q}$ or $\mathbf{G}$) denote vectors/matrices. Then, $\mathbb{E}(\cdot)$, $\mathbb{V}(\cdot, \cdot)$, $\mathbb{D}(\cdot \parallel \cdot)$, $\mathbb{H}(\cdot)$, $\mathbb{I}(\cdot; \cdot)$, $(\cdot)^T$, $\|\cdot\|_2$, $\|\cdot\|_\infty$, $\sum$, and $\sup f(x)$ denote the expectation operator, total variation distance, the relative entropy, the differential entropy, the mutual information, the matrix transpose, the l_2 -norm, the calculation of the infinity norm of a matrix, the summation, and the supremum of the function $f(x)$ over its domain, respectively. Throughout this article, subscripts B and W are used to distinguish parameters associated with the legitimate user (Bob) and the warden (Willie), respectively. Similarly, we use subscript indices j and k to denote the j -th AP and k -th mRIS element, respectively.

II. SYSTEM MODEL

This article considers a dual optical RIS-aided covert MISO-VLC system, which includes a legitimate user (Bob), a warden (Willie), and N_s APs uniformly distributed on the room's ceiling. Each AP relies on an LED array. A direct current-biased Intensity Modulation/Direct Detection (IM/DD) scheme is adopted. Bob and Willie are located on the floor, using Photodiodes (PDs) to receive signals.

In the covert MISO-VLC system, N_s APs transmit optical signals to Bob through N channels. The transmitted signals are represented as $\mathbf{X} = [x_{i,j}]_{N \times N_s}$, where $i = 1, 2, \dots, N$ and $j = 1, 2, \dots, N_s$. The transmitted signal $x_{i,j}$ is subject to the following peak and average optical power constraints,¹ given by

$$\Pr[x_{i,j} > A] = 0, \quad (1)$$

and

$$\sum_{j=1}^{N_s} \mathbb{E}[x_{i,j}] \leq \chi, \quad (2)$$

where A and χ denote the peak optical power and the average optical power, respectively.

An mRIS equipped with N_m reflecting elements is deployed to boost the covert achievable rate for the MISO-VLC system, as depicted in Fig. 1. Meanwhile, an LC-RIS is deployed in front of Bob's PD to enhance the strength of the received signal. Then, the received signal vector at Bob is given by $\mathbf{y}_B = [y_{B,i}]_{N \times 1}$, where $y_{B,i}$ represents the signal received by Bob on the i -th channel, which is expressed as

$$\begin{aligned} y_{B,i} &= \sum_{j=1}^{N_s} \left[\kappa_{B,j}^{\text{LoS}} h_{B,j}^{\text{LoS}} + \sum_{k=0}^{N_m} (\kappa_{B,j,k}^{\text{mRIS}} h_{B,j,k}^{\text{mRIS}} G_{B,j,k}) \right] x_{i,j} + n_{B,i} \\ &= \mathbf{h}_B^T \mathbf{x} + n_{B,i}, \end{aligned} \quad (3)$$

¹Although practical LEDs are subject to individual per-AP power constraints, we adopt a total average optical power constraint to ensure mathematical tractability [30], [31], [32].

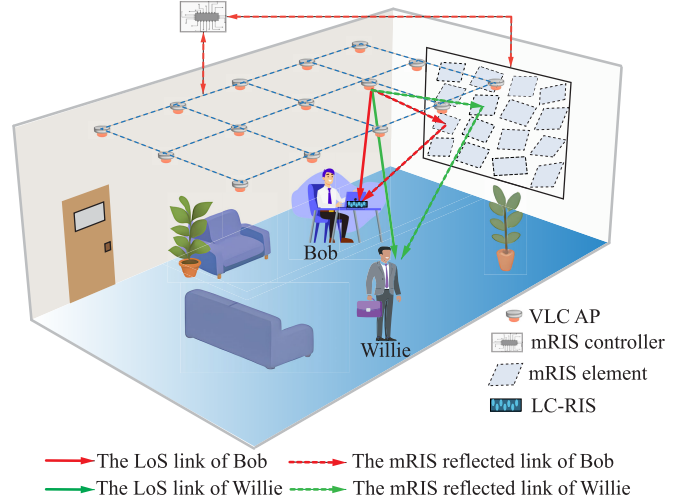


Fig. 1. Dual optical RIS-assisted covert MISO-VLC communication system.

where $\kappa_{B,j}^{\text{LoS}}$ represents the LoS channel gain coefficient provided by the LC-RIS, $h_{B,j}^{\text{LoS}}$ the LoS channel gain from AP j to Bob, $\kappa_{B,j,k}^{\text{mRIS}}$ represents the mRIS channel gain coefficient provided by the LC-RIS, $h_{B,j,k}^{\text{mRIS}}$ represents the mRIS channel gain of the reflection link that is transmitted from AP j , reflected by mirror element k and received at Bob, $\mathbf{h}_B = [h_{B,1}, h_{B,2}, \dots, h_{B,N_s}]^T$, $\mathbf{x} = [x_{i,1}, x_{i,2}, \dots, x_{i,N_s}]^T$. Besides, $G_{B,j,k} \in \{0, 1\}$ indicates AP-mRIS association for Bob, $G_{B,j,k} = 0$ represents the absence of the link transmitted from AP j , reflected by mirror element k , to Bob, i.e., j - k -B link does not exist, $G_{B,j,k} = 1$ represents the existence of the j - k -B link. In addition, the Additive White Gaussian Noise (AWGN) vector at Bob across N channels is represented as $\mathbf{n}_B = [n_{B,i}]_{N \times 1}$, where $n_{B,i} \sim \mathcal{N}(0, \sigma_B^2)$ denotes the noise in the i -th channel, σ_B^2 represents the noise variance at Bob.

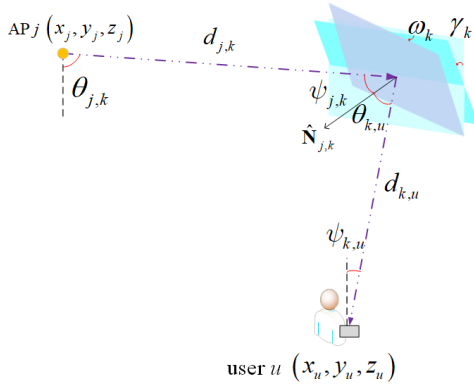
The channel gains of the dual optical RIS-aided covert MISO-VLC system are discussed as follows.

Line-of-sight (LoS) Channel: According to the Lambertian radiation pattern, the LoS channel gain from AP j to user u ($u = B$ for Bob, $u = W$ for Willie) can be expressed as [33]

$$\begin{aligned} h_{u,j}^{\text{LoS}} &= \begin{cases} \frac{(m+1)\tau\xi\vartheta\mu^2 \cos^m(\varphi_{u,j}) \cos(\psi_{u,j})}{2\pi d_{u,j}^2 \sin^2(\Theta_F)}, & 0 \leq \psi_{u,j} \leq \Theta_F, \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (4)$$

where $d_{u,j}$ is the distance from AP j to user u , ϑ is the physical area of the PD, m is the Lambertian coefficient, ξ is the electro-optical conversion efficiency, τ is the responsivity of the PD, $\mathfrak{Z}$ is the gain of the trans-impedance amplifier, μ is the refraction index, $\psi_{u,j}$ and $\varphi_{u,j}$ represent the incident angle and the irradiance angle of the LoS link from AP j to user u , respectively, and Θ_F is half of field-of-view (FoV).

mRIS Channel: The mRIS consists of N_m identical rectangular mirror elements. As shown in Fig. 2, each mirror element can be rotated independently with two degrees of freedom. For mirror element k , the rotation angles include roll angle γ_k and yaw angle ω_k . By intelligently controlling the rotation angles

Fig. 2. Reflection path and rotation diagram for the k -th mRIS element.

of the mirror element k , we are expected to construct a desired optical reflection path to the user u , as shown in Fig. 2. We define $\gamma \in \mathbb{R}^{N_m \times 1}$ and $\omega \in \mathbb{R}^{N_m \times 1}$ as the collections of γ_k and ω_k , where $k = 1, 2, \dots, N_m$.

Considering the reflection characteristics of the mirror element, the path loss of the reflected link is characterized by the additive path loss model.² Then, for an effective specular reflection link where the receiver lies along the reflected direction determined by the orientation of mirror element k , the mRIS channel gain of the link from AP j to user u via mirror element k is expressed by

$$h_{u,j,k}^{\text{mRIS}}(\gamma_k, \omega_k) = \begin{cases} \frac{\rho(m+1)\tau\xi\vartheta\mu^2 dA_k \cos^m(\theta_{j,k})}{2\pi(d_{j,k} + d_{k,u})^2 \sin^2(\Theta_F)} \cos(\psi_{j,k}) \\ \times \cos(\theta_{k,u}) \cos(\psi_{k,u}), & 0 \leq \psi_{k,u} \leq \Theta_F, \\ 0, & \text{otherwise,} \end{cases} \quad (5)$$

where ρ is the mirror element reflection coefficient, $d_{j,k}$ is the distance between AP j and mirror element k , $d_{k,u}$ is the distance between mirror element k and user u , $\theta_{j,k}$ is the angle of irradiance from AP j to mirror element k , $\psi_{j,k}$ is the angle of incidence from AP j to mirror element k , $\theta_{k,u}$ is the angle of irradiance from mirror element k to user u , $\psi_{k,u}$ is the angle of incidence from mirror element k to user u .

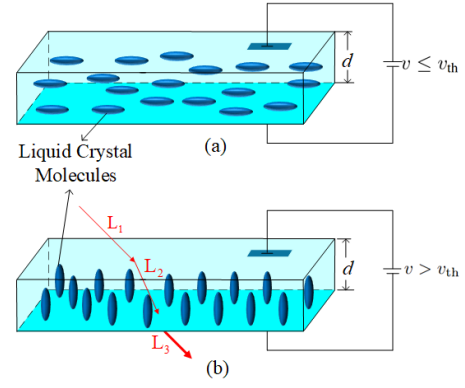
To characterize the two degrees of freedom of the rotation of mirror element k , the cosine of $\theta_{k,u}$ is expressed as [27]

$$\begin{aligned} \cos(\theta_{k,u}) &= \frac{(x_k - x_u)}{d_{k,u}} \sin(\gamma_k) \cos(\omega_k) \\ &+ \frac{(y_k - y_u)}{d_{k,u}} \cos(\gamma_k) \cos(\omega_k) \\ &+ \frac{(z_k - z_u)}{d_{k,u}} \sin(\omega_k), \end{aligned} \quad (6)$$

where (x_k, y_k, z_k) and (x_u, y_u, z_u) are the coordinates of mirror element k and user u , respectively.

Channel Gain Coefficient Provided by LC-RIS-aided Receiver: As shown in Fig. 1, an LC-RIS is introduced to Bob's receiver, which is placed in front of the PD. The LC-RIS is composed of a series of liquid crystal molecules,

²Since the channel gain of diffused light links are so light compared to the LoS links, the diffused light links are not considered in this paper [34].

Fig. 3. (a) The orientation of liquid crystal molecules before applying the voltage v . (b) The orientation of liquid crystal molecules after applying the voltage v .

which can be regarded as elements of the LC-RIS. As shown in Fig. 3, when the externally applied voltage v reaches the critical voltage threshold v_{th} , the tilt angles of the liquid crystal molecules can be changed. By controlling v , we adjust the tilt angles and then tune the refractive index of the LC-RIS to enable light steering and amplification. Specifically, the tilt angle can be expressed as [26]

$$\Delta = \begin{cases} \frac{\pi}{2} - 2 \tan^{-1} \left(\exp \left(-\frac{v - v_{th}}{v_0} \right) \right), & v > v_{th}, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where v_0 is a constant determined by the materials that make up the LC-RIS. Meanwhile, the relationship between the tilt angle Δ and the refractive index η_c of the LC-RIS can be expressed as

$$\eta_c(\Delta) = \sqrt{\left(\frac{\cos^2(\Delta)}{\eta_e^2} + \frac{\sin^2(\Delta)}{\eta_o^2} \right)^{-1}}, \quad (8)$$

where η_e and η_o are the upper bound and the lower bound on the refractive index of the LC-RIS, respectively.

LC-RIS enhances all links incident on the PD, including both LoS and mRIS. To quantify the channel gain provided by LC-RIS, we will use the LoS link as an example below. For the mRIS case, the gain of LC-RIS on mRIS can be obtained by replacing the incident angle accordingly.

Based on Ref. [26], the transmission gain of the LC-RIS can be expressed as

$$\kappa_{u,j}^{\text{LoS}} = \zeta_{u,j}^{\text{LoS}} \ell_{u,j}, \quad (9)$$

where $\zeta_{u,j}^{\text{LoS}}$ represents the transition coefficient of the LC-RIS, $\ell_{u,j}$ represents the amplification coefficient of the LC-RIS.

Furthermore, the quantization of the light reflected from the upper surface of the LC-RIS can be expressed as

$$\begin{aligned} \mathcal{R}_{ac}(\psi_{u,j}, \delta_{u,j}) &= \frac{1}{2} \left(\frac{\eta^2 \cos(\psi_{u,j}) - \cos(\delta_{u,j})}{\eta^2 \cos(\psi_{u,j}) + \cos(\delta_{u,j})} \right)^2 \\ &+ \frac{1}{2} \left(\frac{\cos(\psi_{u,j}) - \cos(\delta_{u,j})}{\cos(\psi_{u,j}) + \cos(\delta_{u,j})} \right)^2, \end{aligned} \quad (10)$$

where the relative refractive index of air with respect to the LC-RIS can be expressed as $\eta = \frac{\eta_c}{\eta_a}$, where the refractive

index of air is represented by η_a . According to Snell's law, we have $\eta_a \sin(\psi_{u,j}) = \eta_c \sin(\delta_{u,j})$. Hence, (10) can be further derived as

$$\mathcal{R}_{ac}(\psi_{u,j}) = \frac{1}{2} \left(\frac{\eta^2 \cos(\psi_{u,j}) - \sqrt{\eta^2 - \sin^2(\psi_{u,j})}}{\eta^2 \cos(\psi_{u,j}) + \sqrt{\eta^2 - \sin^2(\psi_{u,j})}} \right)^2 + \frac{1}{2} \left(\frac{\cos(\psi_{u,j}) - \sqrt{\eta^2 - \sin^2(\psi_{u,j})}}{\cos(\psi_{u,j}) + \sqrt{\eta^2 - \sin^2(\psi_{u,j})}} \right)^2. \quad (11)$$

Assuming that there is no light absorption on the surface of the LC-RIS, the refracted light passing through the LC-RIS can be quantized as

$$\mathcal{T}_{ac}(\psi_{u,j}) = 1 - \mathcal{R}_{ac}(\psi_{u,j}). \quad (12)$$

Similarly, the refracted light from the LC-RIS to air can be quantized as $\mathcal{T}_{ca}(\delta_{u,j}) = 1 - \mathcal{R}_{ca}(\delta_{u,j})$, where $\mathcal{R}_{ca}(\delta_{u,j})$ represents the reflected light and can be quantized as

$$\mathcal{R}_{ca}(\delta_{u,j}) = \frac{1}{2} \left(\frac{\eta^{-2} \cos(\delta_{u,j}) - \sqrt{\eta^{-2} - \sin^2(\delta_{u,j})}}{\eta^{-2} \cos(\delta_{u,j}) + \sqrt{\eta^{-2} - \sin^2(\delta_{u,j})}} \right)^2 + \frac{1}{2} \left(\frac{\cos(\delta_{u,j}) - \sqrt{\eta^{-2} - \sin^2(\delta_{u,j})}}{\cos(\delta_{u,j}) + \sqrt{\eta^{-2} - \sin^2(\delta_{u,j})}} \right)^2. \quad (13)$$

Consequently, the transition coefficient of the LC-RIS $\zeta_{u,j}^{\text{LoS}}$ in (9) can be expressed as

$$\zeta_{u,j}^{\text{LoS}} = \mathcal{T}_{ac}(\psi_{u,j}) \mathcal{T}_{ca}(\delta_{u,j}) = (1 - \mathcal{R}_{ac}(\psi_{u,j})) (1 - \mathcal{R}_{ca}(\delta_{u,j})). \quad (14)$$

Upon the application of an external voltage, the light passing through the LC-RIS can be amplified, as shown Fig. 3. Therefore, the amplification coefficient $\ell_{u,j}$ in (9) can be expressed as

$$\ell_{u,j} = e^{\Gamma d}. \quad (15)$$

Based on the dual-wave dynamic coupling theory, Γ in (15) can be expressed as

$$\Gamma = \frac{2\pi\eta_c^3}{\lambda \cos(\psi_{u,j})} \frac{v}{d} r, \quad (16)$$

where d is the thickness of the LC-RIS, r is the electro-optic coefficient, and λ is the wavelength of the incident light. According to Eqs. (7) and (8), v can be written as a function of η_c , which is expressed as

$$v = v_{\text{th}} - v_0 \ln \left[-\tan \left(\frac{\tan^{-1} \left(\frac{\eta_o \sqrt{(\eta_e^2 - \eta_o^2)(\eta_e^2 - \eta_c^2)}}{\eta_c(\eta_e^2 - \eta_o^2)} \right)}{2} - \frac{\pi}{4} \right) \right]. \quad (17)$$

III. COVERTNESS CONSTRAINT FOR THE MISO-VLC SYSTEM

In this section, we derive the fundamental covertness constraint for the MISO-VLC system. We begin by establishing the detection problem at the warden as a binary hypothesis test. By leveraging the relationship between the total variation distance and the KL divergence via Pinsker's inequality, we transform the covertness requirement into an upper bound on the KL divergence. Subsequently, a general form of the covertness constraint is derived using the Entropy Power Inequality (EPI). Finally, the covertness constraints are specified for two distinct scenarios, which arise from different optimal input signal distributions under different power constraint regimes.

Based on Willie's binary hypothesis testing, let the null hypothesis $\mathcal{H}_0$ indicate that the system is not transmitting information to Bob, while the alternative hypothesis $\mathcal{H}_1$ indicates that the system is transmitting information to Bob. Therefore, the signal vector received by Willie across N channels is given by $\mathbf{y}_W = [y_{W,i}]_{N \times 1}$, where $y_{W,i}$ represents the signal received by Willie on the i -th channel, i.e.,

$$\begin{aligned} \mathcal{H}_0 : y_{W,i} &= n_{W,i}, \\ \mathcal{H}_1 : y_{W,i} &= \sum_{j=1}^{N_s} \left(h_{W,j}^{\text{LoS}} + \sum_{k=1}^{N_m} h_{W,j,k}^{\text{mRIS}} G_{W,j,k} \right) x_{i,j} + n_{W,i} \\ &= \mathbf{h}_W^T \mathbf{x} + n_{W,i}, \end{aligned} \quad (18)$$

where $G_{W,j,k} \in \{0, 1\}$ indicates AP-mRIS association for Willie, the AWGN vector at Willie across N channels is represented as $\mathbf{n}_W = [n_{W,i}]_{N \times 1}$, where $n_{W,i} \sim \mathcal{N}(0, \sigma_W^2)$ denotes the noise in the i -th channel, σ_W^2 represents the noise variance at Willie.

Without loss of generality, this paper considers the transmission of the i -th channel as the representative case. For convenience of representation, we abbreviate $x_{i,j}$, $y_{W,i}$, $y_{B,i}$, $n_{W,i}$ and $n_{B,i}$ as x_j , y_W , y_B , n_W and n_B , respectively.

The detection performance of Willie is typically evaluated using the error probability of detection, i.e., [8]

$$\mathbb{P}_e = \mathbb{P}(\mathcal{H}_0) \mathbb{P}(\mathcal{D}_1 | \mathcal{H}_0) + \mathbb{P}(\mathcal{H}_1) \mathbb{P}(\mathcal{D}_0 | \mathcal{H}_1), \quad (19)$$

where $\mathbb{P}(\mathcal{D}_1 | \mathcal{H}_0)$ and $\mathbb{P}(\mathcal{D}_0 | \mathcal{H}_1)$ represent the false alarm probability and the miss detection probability, respectively, $\mathcal{D}_1$ and $\mathcal{D}_0$ denote the decision in favor of $\mathcal{H}_1$ and $\mathcal{H}_0$, respectively. Since Willie's judgment of the system's transmission state is similar to random guessing [10], [12], we assume that the prior probabilities that the system transmits and does not transmit information are equal, i.e., $\mathbb{P}(\mathcal{H}_0) = \mathbb{P}(\mathcal{H}_1) = 0.5$.

To achieve covert transmission, the system aims for a high $\mathbb{P}_e$ value at Willie's location, meaning that even if Willie uses the optimal detector, the system's transmission cannot be detected. Therefore, the covertness constraint is modeled as

$$\mathbb{P}[\mathcal{D}_1 | \mathcal{H}_0] + \mathbb{P}[\mathcal{D}_0 | \mathcal{H}_1] \geq 1 - \varepsilon, \quad (20)$$

where ε is a small positive constant defining the covertness constraint threshold, such that the communication between APs and Bob maintains ε -indistinguishability from noise observed by the warden.

Based on Ref. [8], Willie's detection error probability can be transformed into

$$\mathbb{P}[\mathcal{D}_1|\mathcal{H}_0] + \mathbb{P}[\mathcal{D}_0|\mathcal{H}_1] = 1 - \mathbb{V}_T(p_1(\mathbf{y}_W), p_0(\mathbf{y}_W)), \quad (21)$$

where $p_0(\mathbf{y}_W)$ denotes the PDF of Willie's observations when the system does not transmit information, $p_1(\mathbf{y}_W)$ denotes the PDF of Willie's observations when the system transmits information. Based on Pinsker's inequality, total variation distance can be transformed into

$$\mathbb{V}_T(p_1(\mathbf{y}_W), p_0(\mathbf{y}_W)) \leq \frac{\ln 2}{2} \sqrt{\mathbb{D}(p_1(\mathbf{y}_W) \| p_0(\mathbf{y}_W))}. \quad (22)$$

According to Eqs. (19) - (22), the system's covertness constraint can be reformulated as

$$\mathbb{D}(p_1(\mathbf{y}_W) \| p_0(\mathbf{y}_W)) \leq \frac{2\varepsilon^2}{\ln 2}. \quad (23)$$

Due to the chain rule of relative entropy, $\mathbb{D}(p_1(\mathbf{y}_W) \| p_0(\mathbf{y}_W))$ can be further expressed as

$$\mathbb{D}(p_1(\mathbf{y}_W) \| p_0(\mathbf{y}_W)) = N\mathbb{D}(p_1(y_W) \| p_0(y_W)). \quad (24)$$

Therefore, the covertness constraint is given by

$$\mathbb{D}(p_1(y_W) \| p_0(y_W)) \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (25)$$

Lemma 1: According to Eq. (25), the covertness constraint of the optical RIS-assisted covert MISO-VLC system can be expressed as

$$\begin{aligned} \mathcal{Z}(A, \mathbf{G}_W) &= \frac{1}{2} \log_2(2\pi\sigma_W^2) + \frac{\mathbb{E}[(\mathbf{h}_W^T \mathbf{x})^2] + \sigma_W^2}{2\sigma_W^2 \ln 2} \\ &\quad - \frac{1}{2} \log_2[2^{2\mathbb{H}(\mathbf{h}_W^T \mathbf{x})} + 2\pi e \sigma_W^2] \leq \frac{2\varepsilon^2}{N \ln 2}, \end{aligned} \quad (26)$$

where $\mathbf{G}_W \in \mathbb{R}^{N_s \times N_m}$ is a matrix containing $G_{W,j,k} \in \{0, 1\}$.

Proof: Based on the definition of relative entropy for continuous variables, $\mathbb{D}(p_1(y_W) \| p_0(y_W))$ in Eq. (25) can be expressed as

$$\begin{aligned} \mathbb{D}(p_1(y_W) \| p_0(y_W)) &= -\mathbb{H}_1(y_W) - \log_2 \left(\frac{1}{\sqrt{2\pi\sigma_W^2}} \right) \\ &\quad + \frac{1}{2\sigma_W^2 \ln 2} \int_{-\infty}^{+\infty} y_W^2 p_1(y_W) dy_W, \end{aligned} \quad (27)$$

where $\mathbb{H}_1(y_W)$ represents the differential entropy of the signal received by Willie when the system is transmitting. According to Eq. (18), $\int_{-\infty}^{+\infty} y_W^2 p_1(y_W) dy_W$ can be derived as

$$\begin{aligned} &\int_{-\infty}^{+\infty} y_W^2 p_1(y_W) dy_W \\ &= \mathbb{E}(y_W^2) \\ &= \mathbb{E} \left(\left(\sum_{j=1}^{N_s} \left[h_{W,j}^{\text{LoS}} + \sum_{k=1}^{N_m} (h_{W,j,k}^{\text{mRIS}} G_{j,k,u}) \right] x_j + n_W \right)^2 \right) \end{aligned}$$

$$\begin{aligned} &= \mathbb{E} \left(\left(\sum_{j=1}^{N_s} \left[h_{W,j}^{\text{LoS}} + \sum_{k=1}^{N_m} (h_{W,j,k}^{\text{mRIS}} G_{j,k,u}) \right] x_j \right)^2 \right) + \sigma_W^2 \\ &= \mathbb{E} \left((\mathbf{h}_W^T \mathbf{x})^2 \right) + \sigma_W^2. \end{aligned} \quad (28)$$

By substituting Eq. (28) into Eq. (27) and using the EPI, an upper bound for $\mathbb{D}(p_1(y_W) \| p_0(y_W))$ can be obtained as

$$\begin{aligned} &\mathbb{D}(p_1(y_W) \| p_0(y_W)) \\ &\leq \frac{1}{2} \log_2(2\pi\sigma_W^2) + \frac{\mathbb{E}[(\mathbf{h}_W^T \mathbf{x})^2] + \sigma_W^2}{2\sigma_W^2 \ln 2} \\ &\quad - \frac{1}{2} \log_2[2^{2\mathbb{H}(\mathbf{h}_W^T \mathbf{x})} + 2\pi e \sigma_W^2]. \end{aligned} \quad (29)$$

By defining the right-hand side of the inequality in (29) as $\mathcal{Z}(A, \mathbf{G}_W)$, and combining it with the requirement from (25), we arrive at the constraint expressed in Lemma 1. ■

Lemma 1 provides a generalized covertness constraint formulation where both $\mathbb{E}[(\mathbf{h}_W^T \mathbf{x})^2]$ and $\mathbb{H}(\mathbf{h}_W^T \mathbf{x})$ are unknown variables, whose closed-form expressions will be derived subsequently in this section.

To derive the unknown variables, we first analyze the detection performance for Willie. Based on information theory, the detection rate can be characterized as

$$\begin{aligned} \mathbb{I}(\mathbf{x}; \mathbf{y}_W) &= N\mathbb{I}(x_1, \dots, x_j; y_W) \\ &= N[\mathbb{H}(\mathbf{h}_W^T \mathbf{x} + n_W) - \mathbb{H}(n_W)]. \end{aligned} \quad (30)$$

Then, according to the EPI, the lower bound of mutual information is given by

$$\begin{aligned} \mathbb{I}(x_1, \dots, x_j; y_W) &= \mathbb{H}(\mathbf{h}_W^T \mathbf{x} + n_W) - \mathbb{H}(n_W) \\ &\geq \frac{1}{2} \log \left(\frac{2^{2\mathbb{H}(\mathbf{h}_W^T \mathbf{x})}}{2\pi e \sigma_W^2} + 1 \right). \end{aligned} \quad (31)$$

Based on the analysis of Eqs. (31) and (26), Willie should render $\mathbb{H}(\mathbf{h}_W^T \mathbf{x})$ as large as possible to achieve better detection performance.

According to Ref. [31], in a MISO-VLC system, $\mathbb{H}(\mathbf{h}_W^T \mathbf{x})$ is governed by the input distribution, which is critically determined by the ratio of the allowed average power to the allowed peak power, denoted as $\mathcal{E} \triangleq \frac{\bar{P}}{P_A}$. Based on Ref. [31], a critical threshold of $\mathcal{E}$ is defined as

$$\mathcal{E}_{\text{th}} \triangleq \frac{1}{2} + \frac{\sum_{j=1}^{N_s} h_{W,j}(j-1)}{\sum_{j=1}^{N_s} h_{W,j}}, \quad (32)$$

where $h_{W,j}$ represents the total channel gain at Willie.

Based on the analysis of $\mathcal{E}$ and $\mathcal{E}_{\text{th}}$, the system operates in two distinct modes: Case I corresponds to a peak-power-limited scenario when $\mathcal{E} \geq \mathcal{E}_{\text{th}}$, while Case II corresponds to a joint peak-and-average-power-limited scenario when $0 < \mathcal{E} < \mathcal{E}_{\text{th}}$. In this paper, for both cases, we derive closed-form expressions for $\mathbb{E}[(\mathbf{h}_W^T \mathbf{x})^2]$ and $\mathbb{H}(\mathbf{h}_W^T \mathbf{x})$, formally establishing the corresponding covertness constraints, which are given in Theorem 1 and Theorem 2, respectively.

Theorem 1: In Case I, when $\Xi \geq \Xi_{\text{th}}$, the covertness constraint is expressed as

$$\mathcal{Z}_I(A, \mathbf{G}_W) = \frac{1}{2} \log_2(2\pi\sigma_W^2) + \frac{\frac{1}{3}(\|\mathbf{h}_W^T\|_\infty A)^2 + \sigma_W^2}{2\sigma_W^2 \ln 2} - \frac{1}{2} \log_2 \left[(\|\mathbf{h}_W^T\|_\infty A)^2 + 2\pi e \sigma_W^2 \right] \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (33)$$

Proof: See Appendix A. ■

To analyze case II, we first denote

$$s_{u,0} \triangleq 0, \quad (34)$$

and

$$s_{u,j} \triangleq \sum_{j'=1}^j h_{u,j'}, \quad (35)$$

where $u \in \{B, W\}$, $h_{u,j'}$ represents the total channel gain from AP j' to user u in the dual optical RIS-aided VLC system. Then, we have the following theorem.

Theorem 2: In Case II, when $0 < \Xi < \Xi_{\text{th}}$, the covertness constraint is expressed as

$$\begin{aligned} \mathcal{Z}_{II}(A, \mathbf{G}_W) &= \frac{1}{2\sigma_W^2 \ln 2} \sum_{j=1}^{N_s} p_{W,j} \frac{\mu A^2}{1 - e^{-\mu(\lambda)}} (t_1 + t_2 + t_3) \\ &\quad - \frac{1}{2 \ln 2} \ln \left[1 + \frac{(s_{W,N_s} A)^2 \cdot 2^{2\mathcal{V}_W}}{2\pi e \sigma_W^2} \right] \\ &\leq \frac{2\varepsilon^2}{N \ln 2}, \end{aligned} \quad (36)$$

where $t_1 = s_{W,j-1}^2 \frac{1-e^{-\mu}}{\mu}$, $t_2 = 2s_{W,j-1} h_{W,j} \frac{1-(1+\mu)e^{-\mu}}{\mu^2}$, $t_3 = h_{W,j}^2 \frac{2-2(1+\mu+\frac{\mu^2}{2})e^{-\mu}}{\mu^3}$ and

$$\begin{aligned} \mathcal{V}_W &= \sup_{\gamma \in (\max\{0, \frac{1}{2} + \Xi - \Xi_{\text{th}}\}, \min\{\frac{1}{2}, \Xi\})} \left\{ 1 - \log \frac{\mu(\gamma)}{1 - e^{-\mu(\gamma)}} \right. \\ &\quad \left. - \frac{\mu(\gamma) e^{-\mu(\gamma)}}{1 - e^{-\mu(\gamma)}} - \mathbb{D} \left(\mathbf{p}_W \parallel \frac{\mathbf{h}_W}{s_{W,N_s}} \right) \right\}, \end{aligned} \quad (37)$$

where $\mu(\gamma)$ is the unique positive solution to

$$\frac{1}{\mu} - \frac{e^{-\mu}}{1 - e^{-\mu}} = \gamma, \quad (38)$$

and $\mathbf{p}_W = [p_{W,j}]_{N_s \times 1}^T$, $p_{W,j} = \frac{h_{W,j} a^j}{\sum_{k=1}^{N_s} h_{W,k} a^k}$ represent selection probability, where a is the unique positive solution to

$$\frac{\sum_{k=1}^{N_s} h_{W,k} k a^k}{\sum_{j=1}^{N_s} h_{W,j} a^j} = \Xi - \gamma + 1. \quad (39)$$

Proof: See Appendix B. ■

IV. OPTIMIZATION OF DUAL OPTICAL RIS-ASSISTED COVERT MISO-VLC SYSTEM

In this section, we first formulate an optimization problem to maximize the achievable covert rate for the dual optical RIS-aided VLC system. Then, through analysis of the physical characteristics and geometric principles governing optical transmission, we decouple the problem into three subproblems. Leveraging the principle of AP proximity selection and the reflected spot searching, we propose a low-complexity GDDO algorithm to solve the subproblems.

A. Problem Formulation

To evaluate the system performance under covert constraints, the achievable rate at Bob is derived for two cases following the method in Ref. [31]. Particularly, when $\Xi \geq \Xi_{\text{th}}$, the achievable rate at Bob is derived as

$$R_{B,I} = \frac{1}{2} \log \left(1 + \frac{s_{B,N_s}^2 A^2}{2\pi e \sigma_B^2} \right). \quad (40)$$

Besides, when $0 < \Xi < \Xi_{\text{th}}$, the achievable rate at Bob is derived as

$$R_{B,II} = \frac{1}{2} \log \left(1 + \frac{s_{B,N_s}^2 A^2}{2\pi e \sigma_B^2} e^{2\mathcal{V}_B} \right), \quad (41)$$

where $\mathcal{V}_B$ is similar to $\mathcal{V}_W$ in Eq. (37). For the reader's convenience, we summarize the key results of case I and case II in Table I.

To enhance the system's covert performance, we introduce mRIS and LC-RIS to the MISO-VLC system. For the mRIS, we aim to adjust the rotation of mirror elements to control the reflection of VLC links. We define the AP-mRIS association through a three-dimensional matrix $\mathbf{G} \in \mathbb{R}^{N_s \times N_m \times 2}$ with elements $G_{u,j,k}$, where $u \in \{B, E\}$, $j = 1, \dots, N_s$ and $k = 1, \dots, N_m$. Since the distance between the different APs is much larger than the size of each mirror element in most realistic scenarios, we assume that the k -th mRIS element can be assigned to no more than one AP. Hence, $\mathbf{G}$ has a row summation constraint, which is expressed as

$$\sum_{j=1}^{N_s} G_{u,j,k} \leq 1. \quad (42)$$

Similarly, we also assume that each mRIS element can only reflect signals to a single user, which results in the constraint

$$\sum_{u \in \{B, W\}} G_{u,j,k} \leq 1. \quad (43)$$

In addition, considering the rotation ranges provided by the servo electric cylinder, the roll angle and yaw angle for the k -th mRIS element are constrained as

$$-\frac{\pi}{2} \leq \gamma_k \leq \frac{\pi}{2}, \quad (44)$$

and

$$-\frac{\pi}{2} \leq \omega_k \leq \frac{\pi}{2}, \quad (45)$$

respectively.

For the LC-RIS, we aim to adjust the refractive index to control the refraction of VLC links. Considering the physical

TABLE I
SUMMARY OF COVERT COMMUNICATION METRICS UNDER DIFFERENT POWER CONSTRAINTS

Case	I	II
Intensity Constraint	Peak power	Both peak and average power
Range of Ξ	$\Xi \geq \Xi_{\text{th}}$	$0 < \Xi < \Xi_{\text{th}}$
Covertiness Constraint	$\mathcal{Z}_I(A, \mathbf{G}_W) \leq \frac{2\epsilon^2}{N \ln 2}$	$\mathcal{Z}_{II}(A, \mathbf{G}_W) \leq \frac{2\epsilon^2}{N \ln 2}$
Achievable Rate [31]	$R_{B,I} = \frac{1}{2} \log \left(1 + \frac{s_{B,N_s}^2 A^2}{2\pi e \sigma_B^2} \right)$	$R_{B,II} = \frac{1}{2} \log \left(1 + \frac{s_{B,N_s}^2 A^2}{2\pi e \sigma_B^2} e^{2\nu_B} \right)$
Input Distribution	The input signal $\mathbf{h}_W^T \mathbf{x}$ follows a uniform distribution over the interval $(0, \ \mathbf{h}_W^T\ _\infty A]$.	The input distribution follows a truncated exponential distribution as given in (101).

stability of the liquid crystal material, the controllable refractive index η_c is constrained as $\eta_o \leq \eta_c \leq \eta_e$.

Based on the covertness constraints in Theorem 1 and Theorem 2, we formulate an achievable rate maximization problem for the dual-optical-RIS-assisted covert MISO-VLC system, which is expressed as

$$(P) \max_{A, \mathbf{G}, \eta_c, \gamma, \omega} R_{B,n}(A, \eta_c, \gamma, \omega, \mathbf{G}_B) \quad (46)$$

$$\text{s.t. } \mathcal{Z}_n(A, \mathbf{G}_W) \leq \frac{2\epsilon^2}{N \ln 2}, \quad (47)$$

$$\eta_o \leq \eta_c \leq \eta_e, \quad (48)$$

$$-\frac{\pi}{2} \leq \gamma_k \leq \frac{\pi}{2}, \quad (49)$$

$$-\frac{\pi}{2} \leq \omega_k \leq \frac{\pi}{2}, \quad (50)$$

$$\sum_{j=1}^{N_s} G_{u,j,k} \leq 1, \quad (51)$$

$$\sum_{u \in \{B, W\}} G_{u,j,k} \leq 1, \quad (52)$$

where $j = 1, \dots, N_s, k = 1, \dots, N_m, n \in \{I, II\}$, where $n = I$ and $n = II$ denote Case I and Case II, respectively.

B. Proposed GDDO Algorithm

The formulated problem (P) is a non-convex large-scale MINLP problem. Typically, finding the optimal solution to a combinatorial optimization problem (P) is intractable, whose complexity is at least $\mathcal{O}\left((N_s + 1)^{N_m} \cdot (L_\gamma L_\omega)^{N_m}\right)$, where L_γ and L_ω are the quantization levels for the rotation angles. To solve this problem, we propose a GDDO algorithm, which leverages the principle of AP proximity selection for AP-mRIS association, determines optimal mRIS configurations via reflective spot searching, and optimizes LC-RIS parameters through the gradient descent method. Detailed procedures of the GDDO algorithm are provided as follows.

Optimization of the Configuration of mRIS: When the transmission power and LC-RIS refractive index are fixed, problem (P) reduces to an mRIS optimization subproblem (P1) for configuring the element alignment matrix $\mathbf{G}$ and rotation of the mRIS element, which can be expressed as

$$(P1) \max_{\mathbf{G}, \gamma, \omega} R_{B,n}(\gamma, \omega, \mathbf{G}_B | A, \eta_c) \quad (53)$$

$$\text{s.t. } \mathcal{Z}_n(A, \mathbf{G}_W) \leq \frac{2\epsilon^2}{N \ln 2}, \quad (54)$$

$$-\frac{\pi}{2} \leq \gamma_k \leq \frac{\pi}{2}, \quad (55)$$

$$-\frac{\pi}{2} \leq \omega_k \leq \frac{\pi}{2}, \quad (56)$$

$$\sum_{j=1}^{N_s} G_{u,j,k} \leq 1, \quad (57)$$

$$\sum_{u \in \{B, W\}} G_{u,j,k} \leq 1. \quad (58)$$

To maximize the achievable rate for the covert VLC system, the mirror array should serve Bob only. Therefore, by using geometric optics, for AP j and mirror element k , we have

$$\mathbf{G}_W^* = [G_{W,j,k}]_{N_s \times N_m} = [0]_{N_s \times N_m}. \quad (59)$$

Physically, this assumption (59) is justified by the high directionality of mRIS reflections, which form a narrow beam strictly focused on Bob. Furthermore, even accounting for diffuse reflections where each mRIS element acts as a virtual light source, the signals reaching Willie via wall scattering suffer significant attenuation, rendering them negligible compared to the main link.

Based on constraint (59), constraints (57) and (58) can be simplified to only limit Bob's AP association, i.e.,

$$\sum_{j=1}^{N_s} G_{B,j,k} \leq 1, \quad (60)$$

thereby we can transform (P1) into

$$(P1-1) \max_{\mathbf{G}_B, \gamma, \omega} R_{B,n}(\gamma, \omega, \mathbf{G}_B | A, \eta_c) \quad (61)$$

$$\text{s.t. } -\frac{\pi}{2} \leq \gamma_k \leq \frac{\pi}{2}, \quad (62)$$

$$-\frac{\pi}{2} \leq \omega_k \leq \frac{\pi}{2}, \quad (63)$$

$$\sum_{j=1}^{N_s} G_{B,j,k} \leq 1. \quad (64)$$

Problem (P1-1) is still a complex high-dimensional problem. To address (P1-1) efficiently, we leverage the geometric characteristics of the optical channel. In particular, with APs uniformly distributed on the ceiling, the distances from different APs to the same mRIS element vary significantly.

Governed by the Lambertian radiant pattern, the optical channel gain is highly sensitive to this transmission distance, resulting in a substantial path loss disparity between the nearest AP and the second-nearest one.

Motivated by this path loss characteristic, we propose a low-complexity AP proximity selection strategy. Specifically, when Bob is moving around the room, all of the mirror elements in the mRIS will associate with the closest AP, i.e., the j^* -th AP. At this time, the optimal solution of $\mathbf{G}_B$, denoted as $\mathbf{G}_B^*$, can be expressed as

$$\begin{cases} \sum_{k=1}^{N_m} G_{B,j^*,k} = N_m, & \text{if } j = j^*, \\ 0, & \text{otherwise.} \end{cases} \quad (65)$$

Therefore, subproblem (P1) can be further transformed into

$$(P1-2) \max_{\gamma, \omega} R_{B,n}(\gamma, \omega | A, \eta_c, \mathbf{G}_B^*) \quad (66)$$

$$\text{s.t.} \quad \text{s.t.} \quad -\frac{\pi}{2} \leq \gamma_k \leq \frac{\pi}{2}, \quad (67)$$

$$-\frac{\pi}{2} \leq \omega_k \leq \frac{\pi}{2}. \quad (68)$$

Lemma 2: The position of the reflected light spot and the rotation angles of the mRIS element exhibit a one-to-one geometric mapping.

Proof: See Appendix C. ■

According to Lemma 2, considering the one-to-one geometric mapping relationship between the position of the reflected light spot on the receiving plane and the rotation angles, the search for the optimal rotation angles γ and ω is transformed to finding the optimal position of the reflected spot $\mathbf{Q}$, i.e.,

$$(P1-3) \max_{\mathbf{Q}} R_{B,n}(\mathbf{Q} | A, \eta_c, \mathbf{G}_B^*) \quad (69)$$

$$\text{s.t.} \quad 0 \leq x_q \leq x_r \quad (70)$$

$$0 \leq y_q \leq y_r \quad (71)$$

where $\mathbf{Q} = [x_q, y_q, z_q]$ represents the position of the reflected spot.

To enhance the reception for Bob, the optimal position of the reflected spot is Bob's position. Therefore, the solution of (P1-3) can be expressed as $\mathbf{Q}^* = [x_B, y_B, z_B]$. Accordingly, the obtained mirror orientations correspond to reflected links aligned with Bob, rather than to the evaluation of reflected power at arbitrary off-axis observation points.

Based on Lemma 2, the corresponding optimal solutions γ^* and ω^* in subproblem (P1-2) can be calculated by

$$\begin{cases} \omega_k^* = \sin^{-1} \left(\hat{\mathbf{N}}_{j^*,k}^T(\mathbf{Q}^*) \mathbf{e}_3 \right), \\ \gamma_k^* = \sin^{-1} \left(\frac{\hat{\mathbf{N}}_{j^*,k}^T(\mathbf{Q}^*) \mathbf{e}_1}{\cos(\omega_k^*)} \right), \end{cases} \quad (72)$$

where $\mathbf{e}_1 = [1, 0, 0]^T$, $\mathbf{e}_3 = [0, 0, 1]^T$, $\hat{\mathbf{N}}_{j^*,k}(\mathbf{Q}^*)$ represents the normal vector of the mirror element k after rotation

towards $\mathbf{Q}^*$, and according to Snell's law of reflection, it can be expressed as

$$\hat{\mathbf{N}}_{j^*,k}(\mathbf{Q}^*) = \frac{\frac{\mathbf{S}_{j^*} - \mathbf{R}_k}{\|\mathbf{S}_{j^*} - \mathbf{R}_k\|_2} + \frac{\mathbf{Q}^* - \mathbf{R}_k}{\|\mathbf{Q}^* - \mathbf{R}_k\|_2}}{\left\| \frac{\mathbf{S}_{j^*} - \mathbf{R}_k}{\|\mathbf{S}_{j^*} - \mathbf{R}_k\|_2} + \frac{\mathbf{Q}^* - \mathbf{R}_k}{\|\mathbf{Q}^* - \mathbf{R}_k\|_2} \right\|_2}, \quad (73)$$

where $\mathbf{S}_{j^*}$ represents the position of AP j^* .

Consequently, once the optimal reflection spots and AP-mRIS associations are determined, γ^* , ω^* are uniquely defined via (72) and (73). Leveraging this one-to-one geometric mapping, we transform the search in the angular space into a search for reflection spot positions, thereby significantly simplifying the optimization.

Optimization of the Configuration of LC-RIS: By AP proximity selection and Lemma 2, we obtain γ^* , ω^* , $\mathbf{G}_B^*$ and $\mathbf{G}_W^*$. Then, with a fixed transmission power, problem (P) reduces to an LC-RIS optimization subproblem (P2) for configuring the refractive index which can be expressed as

$$(P2) \max_{\eta_c} R_{B,n}(\eta_c | A, \gamma^*, \omega^*, \mathbf{G}_B^*) \quad (74)$$

$$\text{s.t.} \quad \eta_o \leq \eta_c \leq \eta_e. \quad (75)$$

From Eq. (3), since the refractive index of the LC-RIS solely affects the channel gain, maximizing covert rate is equivalent to maximizing the channel gain at Bob. Therefore, problem (P2) can be reformulated as

$$(P2-1) \max_{\eta_c} \sum_{j=1}^{N_s} (\kappa_{B,j}^{\text{LoS}} h_{B,j}^{\text{LoS}}) + \sum_{k=1}^{N_m} (\kappa_{B,j^*,k}^{\text{mRIS}} h_{B,j^*,k}^{\text{mRIS}} G_{B,j^*,k}) \quad (76)$$

$$\text{s.t.} \quad \eta_o \leq \eta_c \leq \eta_e. \quad (77)$$

Since the LoS channel and mRIS channel is independent of η_c , the concavity of the objective function in (76) is exclusively introduced by the LC-RIS transmission gain. The second-order derivative verification provided in Appendix D demonstrates that (76) represents a convex optimization problem that can be efficiently solved by the gradient descent method.

Optimization of the Transmission Peak Power: With the optimal γ^* , ω^* , $\mathbf{G}_B^*$, $\mathbf{G}_W^*$ and η_c^* , subproblem (P3) can be expressed as

$$(P3) \max_A R_{B,n}(A | \eta_c^*, \gamma^*, \omega^*, \mathbf{G}_B^*) \quad (78)$$

$$\text{s.t.} \quad \mathcal{Z}_n(A, \mathbf{G}_W^*) \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (79)$$

When $n = \text{I}$, corresponding to a peak-power-limited scenario (i.e., $\varepsilon \geq \varepsilon_{\text{th}}$), subproblem (P3) can be further expressed as

$$(P3\text{I}) \max_A R_{B,\text{I}}(A | \eta_c^*, \gamma^*, \omega^*, \mathbf{G}_B^*) \quad (80)$$

$$\text{s.t.} \quad \mathcal{Z}_{\text{I}}(A, \mathbf{G}_W^*) \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (81)$$

According to Theorem 1, the covertness constraint $\mathcal{Z}_I(A, \mathbf{G}_W^*) \leq \frac{2\varepsilon^2}{N \ln 2}$ is arranged as

$$\frac{(\|\mathbf{h}_W^T\|_\infty A)^2}{6\sigma_W^2 \ln 2} - \frac{1}{2 \ln 2} \ln \left[\frac{(\|\mathbf{h}_W^T\|_\infty A)^2}{2\pi e \sigma_W^2} + 1 \right] \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (82)$$

By using the inequality $\ln(1+x) \geq x - \frac{1}{2}x^2$, we have

$$\begin{aligned} & \frac{(\|\mathbf{h}_W^T\|_\infty A)^2}{6\sigma_W^2 \ln 2} \\ & - \frac{1}{2 \ln 2} \left[\frac{(\|\mathbf{h}_W^T\|_\infty A)^2}{2\pi e \sigma_W^2} - \frac{1}{2} \left(\frac{(\|\mathbf{h}_W^T\|_\infty A)^2}{2\pi e \sigma_W^2} \right)^2 \right] \\ & \leq \frac{2\varepsilon^2}{N \ln 2}. \end{aligned} \quad (83)$$

Solving Eq. (83), we further obtain

$$\begin{aligned} A & \leq \sqrt{\frac{8\pi^2 e^2 \sigma_W^2}{\|\mathbf{h}_W^T\|_\infty^2} \left[\sqrt{\left(\frac{1}{6} - \frac{1}{4\pi e} \right)^2 + \frac{\varepsilon^2}{4\pi^2 e^2 N}} - \left(\frac{1}{6} - \frac{1}{4\pi e} \right) \right]} \\ & \triangleq A_I^*. \end{aligned} \quad (84)$$

Equation (84) characterizes the condition under which $\Xi \geq \Xi_{th}$, defining the maximum peak power that the APs can utilize to satisfy the covertness constraint (81). Therefore, A_I^* can be regarded as one of the feasible solutions to subproblem (P3I).

Remark 1: The inequality $\ln(1+x) \geq x - \frac{1}{2}x^2$ is used to transform the covertness constraint into a tractable form. This approximation, however, shrinks the feasible region of the original problem, leading to a conservative upper bound A_I^* in (84). Consequently, the obtained peak power is guaranteed to satisfy the covertness constraint, but it may be suboptimal compared to the global optimum of the original problem.

When $n = \text{II}$, corresponding to the joint peak-and-average-power-limited scenario (i.e., $0 < \Xi < \Xi_{th}$), subproblem (P3) can be further expressed as

$$(\text{P3II}) \max_A R_{B,\text{II}}(A|\eta_c^*, \gamma^*, \omega^*, \mathbf{G}_B^*) \quad (85)$$

$$\text{s.t.} \quad \mathcal{Z}_{\text{II}}(A, \mathbf{G}_W^*) \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (86)$$

To establish a tractable form of the covertness constraint (2), we apply the inequality $\ln(1+x) \geq x - \frac{1}{2}x^2$ to transform the constraint into a biquadratic inequality, which is expressed as

$$C_4 A^4 + C_3 A^3 + C_2 A^2 + C_1 A + C_0 \leq 0, \quad (87)$$

where the coefficients are identified as

$$C_4 = \frac{s_{W,N_s}^4 2^{4v_W}}{16\pi^2 e^2 \sigma_W^4 \ln 2}, \quad (88)$$

$$C_3 = 0, \quad (89)$$

$$\begin{aligned} C_2 &= \frac{1}{2\sigma_W^2 \ln 2} \sum_{j=1}^{N_s} p_{W,j} \frac{\mu A^2}{1 - e^{-\mu}} (t_1 + t_2 + t_3) \\ & - \frac{s_{W,N_s}^2 2^{2v_W}}{4\pi e \sigma_W^2 \ln 2}, \end{aligned} \quad (90)$$

$$C_1 = 0, \quad (91)$$

Algorithm 1 Geometrically-Driven Decoupling Optimization (GDDO) Algorithm

Input: Bob's position $\mathbf{P}_B = (x_B, y_B, z_B)$, Willie's position $\mathbf{P}_W = (x_W, y_W, z_W)$ and N_s, N_m .

Output: Optimal solution $\mathbf{G}^*, \gamma^*, \omega^*, \eta_c^*, A^*$.

- 1: Set $\mathbf{G}_W^* \leftarrow [\mathbf{0}]_{N_s \times N_m}$.
- 2: $j^* \leftarrow$ AP proximity selection.
- 3: Set $\mathbf{G}_B^*$ based on j^* according to Eq. (65).
- 4: Set target spot $\mathbf{Q}^* \leftarrow \mathbf{P}_B$.
- 5: **for** $k = 1$ to N_m **do**
- 6: Calculate optimal angles (γ_k^*, ω_k^*) from $\mathbf{Q}^*$ via Eqs. (72), (73).
- 7: **end for**
- Initialize:
- 8: learning rate g , max iterations $T_{\max}$, tolerance ϵ_{th} .
- 9: Set initial $\eta_c^{(0)}$.
- 10: **for** $t = 1$ to $T_{\max}$ **do**
- 11: Update via gradient ascent: $\eta_c^{(t+1)} \leftarrow \eta_c^{(t)} + g \cdot \frac{\partial h_B(\eta_c)}{\partial \eta_c}$.
- 12: **if** $|h_B(\eta_c^{(t+1)}) - h_B(\eta_c^{(t)})| < \epsilon_{th}$ **then**
- 13: **break**
- 14: **end if**
- 15: **end for**
- 16: Set optimal value: $\eta_c^* \leftarrow \eta_c^{(t+1)}$.
- 17: Calculate channel gains $\mathbf{h}_B$ and $\mathbf{h}_W$ with the obtained parameters: $\mathbf{G}^*, \gamma^*, \omega^*, \eta_c^*, \mathbf{P}_W$ and $\mathbf{P}_B$.
- 18: Calculate A_I^* via Eq. (84) or A_{II}^* via Eq. (94).
- 19: **return** $\mathbf{G}^*, \gamma^*, \omega^*, \eta_c^*, A_I^*$ or A_{II}^* .

and

$$C_0 = -\frac{2\varepsilon^2}{N \ln 2}. \quad (92)$$

By treating A^2 as a single variable and applying the quadratic formula, we obtain

$$0 < A^2 \leq \frac{-C_2 + \sqrt{C_2^2 - 4C_4 C_0}}{2C_4} > 0. \quad (93)$$

Furthermore, we obtain the upper bound on the peak power as

$$A \leq \sqrt{\frac{-C_2 + \sqrt{C_2^2 - 4C_4 C_0}}{2C_4}} \triangleq A_{\text{II}}^*. \quad (94)$$

Equation (94) characterizes the condition under $0 < \Xi < \Xi_{th}$, defining the maximum peak power that the APs can utilize to satisfy the covertness constraint (86). Therefore, A_{II}^* can be regarded as one of the feasible solutions to subproblem (P3II).

Remark 2: Similar to Case I, the inequality $\ln(1+x) \geq x - \frac{1}{2}x^2$ is applied to obtain a tractable biquadratic inequality. This approximation also results in a conservative upper bound A_{II}^* in (94), ensuring the satisfaction of the covertness constraint at the possible cost of optimality.

The key steps of the GDDO algorithm are summarized in Algorithm 1. To assess the feasibility of the proposed approach, we discuss the computational complexity of the proposed GDDO algorithm. Firstly, based on the nearest-neighbor principle, the AP-mRIS association involves traversing N_s APs to determine the optimal connection, which requires $\mathcal{O}(N_s)$

TABLE II
SIMULATION PARAMETERS

Parameter	Value
FoV of PD, Θ_F	80°
Effective physical area of PD, ϑ	1 cm^2
Voltage constant of LC-RIS, v_0	0.8 V
Threshold voltage of LC-RIS, v_{th}	1.34 V
Electro-optical conversion coefficient, r	12 nm/V
Lower constraint on refractive index, η_o	1.5
Upper constraint on refractive index, η_e	1.7
Responsivity of the PD, τ	0.54 A/W
Refraction index of lenses in PD, μ	1.5
Refractive index of air, η_a	1
Reflection efficiency of mirrors, ρ	0.9
mRIS element power consumption, P_m	100 mW [26]
System's intrinsic power, $P_{intrinsic}$	9062.4 mW [26]
LC-RIS power consumption, P_{LC}	320 mW [26]

operations. Also, following AP selection, the computation of precise channel gains involves distances, dot products, and trigonometric operations, resulting in complexities of $\mathcal{O}(N_s)$ for the LoS links and $\mathcal{O}(N_m)$ for the NLoS links. Furthermore, the LC-RIS optimization phase involves T_{max} iterations; in each step, evaluating the objective function and gradient across all LoS and mRIS paths incurs a cost of $\mathcal{O}(N_s + N_m)$, leading to a total of $\mathcal{O}(T_{max}(N_s + N_m))$ operations.

Consequently, the total computational complexity of the GDDO algorithm is expressed as $\mathcal{O}(N_s + N_m + T_{max}(N_s + N_m))$. This analysis demonstrates that the complexity scales linearly with the number of RIS elements N_m , validating the efficiency of the proposed method for large-scale implementations. Crucially, unlike RF-RIS systems that suffer from prohibitive channel estimation overhead, our geometry-driven approach relies solely on position coordinates, thereby effectively circumventing the need for frequent, element-wise Channel State Information (CSI) feedback.

V. NUMERICAL RESULTS

This section provides simulation results to show the feasibility of the dual optical RIS-aided MISO-VLC system and validate the proposed GDDO algorithm. The size of the room is $6 \text{ m} \times 6 \text{ m} \times 3 \text{ m}$. The main simulation parameters are shown in Table II.

To evaluate the tightness of the EPI-based upper bound and approximated upper bound, we employ Monte Carlo simulations. First, without loss of generality, we consider Case I and derive the exact analytical expression for the probability density function (PDF) of the signal received at Willie, given as

$$p_0(y_W) = \frac{1}{\sqrt{2\pi}\sigma_W} e^{-\frac{y_W^2}{2\sigma_W^2}}, \quad (95)$$

and

$$p_1(y_W) = \frac{1}{\|\mathbf{h}_W^T\|_\infty A} \left[Q\left(-\frac{y_W}{\sigma_W}\right) - Q\left(\frac{\|\mathbf{h}_W^T\|_\infty A - y_W}{\sigma_W}\right) \right], \quad (96)$$

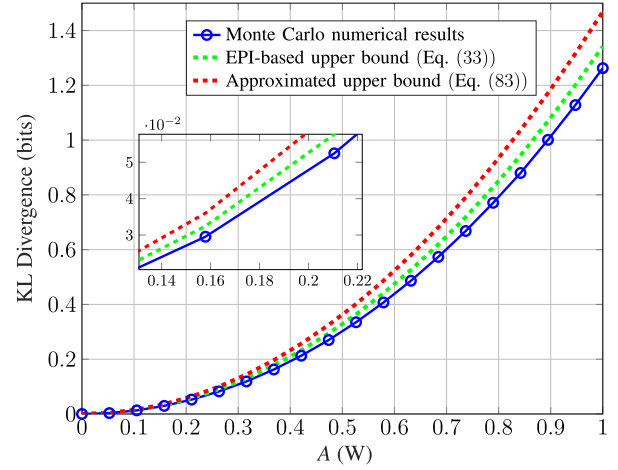


Fig. 4. Comparison of the Monte Carlo numerical results with the EPI-based upper bound and the approximation upper bound in Eq. (83) versus the peak power A .

where $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x e^{-t^2/2} dt$ denotes the Gaussian Q-function.

Proof: See Appendix E. ■

With the exact expression for $p_0(y_W)$ and $p_1(y_W)$, we conducted extensive Monte Carlo simulations to calculate the numerical KL divergence. The results are shown in Fig. 4, which compares the Monte Carlo numerical results, the EPI-based upper bound, and the approximation upper bound in Eq. (83) versus the peak power A . It is observed that the EPI-based upper bound consistently lies above the Monte Carlo numerical results, validating the correctness of our theoretical analysis. Furthermore, the approximation upper bound is derived by relaxing the EPI-based bound via the $\ln(1+x)$ inequality, serving as a further upper bound that lies strictly above the EPI curve. Notably, the approximation upper bound tightly converges with the Monte Carlo numerical results in the low transmit power regime. This justifies utilizing the EPI and the $\ln(1+x)$ inequality to formulate a tractable upper bound, which simplifies the optimization process while yielding near-optimal solutions.

Fig. 5 depicts the covert transmission rates of the mRIS-assisted covert MISO-VLC system at various indoor positions in case I, with mRIS element counts $N_m = 0, 25, 49, 100$. In Fig. 5, four APs are uniformly distributed on the ceiling at positions (1, 1, 3), (5, 1, 3), (5, 5, 3), and (1, 5, 3). Specifically, Fig. 5(a)-Fig. 5(d) are top views that are colored based on the value of covert transmission rate, where the region with higher brightness represents a larger rate. As shown in Fig. 5 (a), under covertness constraints, lower transmit power and spaced AP locations yield higher covert transmission rates beneath APs than elsewhere in the room. The introduction of mRIS enhances covert transmission rates throughout the room, as shown in Fig. 5 (b)-Fig. 5 (d). With the number of mirror elements increasing from 25 to 100, both the transmission rates and covert coverage area gradually expand, demonstrating progressive improvements relative to the no-mRIS baseline. In contrast, Fig. 6 presents a scenario with reduced AP spacing, where four APs are positioned at (2, 2, 3), (4, 2, 3),

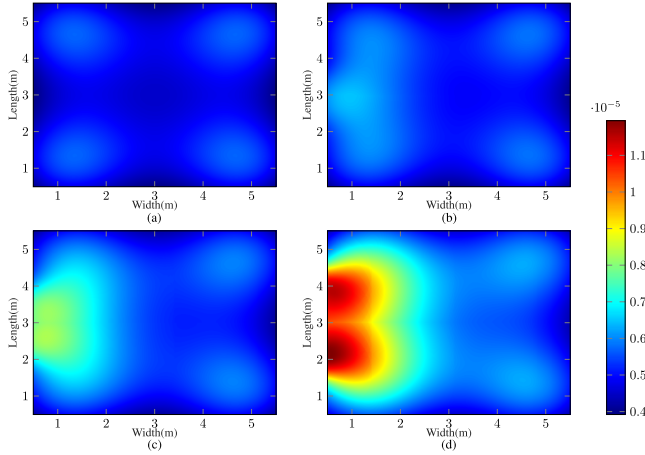


Fig. 5. Case I: Covert transmission rate of the mRIS-assisted covert MISO-VLC at different positions in the room versus the number of mRIS elements, N_m , with the spacing between mRIS elements being 0.2m. (a) $N_m=0$. (b) $N_m=25$. (c) $N_m=49$. (d) $N_m=100$.

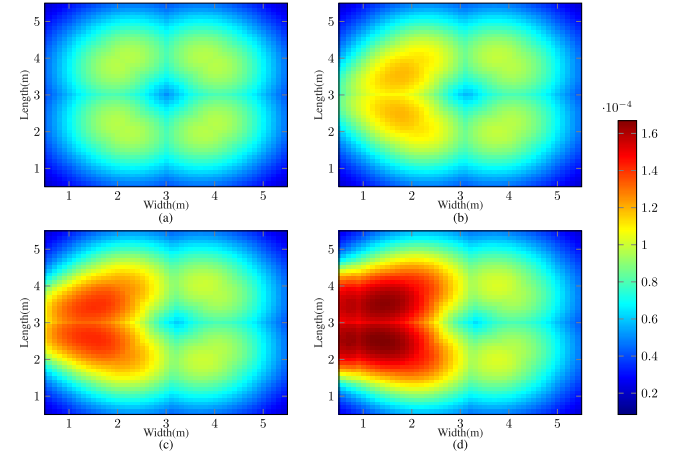


Fig. 7. Case II: Covert transmission rate of the mRIS-assisted covert MISO-VLC at different positions in the room versus the number of mRIS elements, N_m , with the spacing between mRIS elements being 0.2m. (a) $N_m=0$. (b) $N_m=25$. (c) $N_m=49$. (d) $N_m=100$.

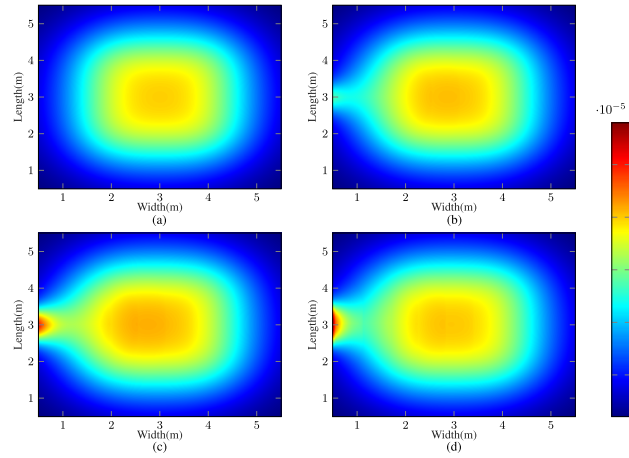


Fig. 6. Case I: Covert transmission rate of the mRIS-assisted covert MISO-VLC at different positions in the room versus the number of mRIS elements, N_m , with the spacing between mRIS elements being 0.1m. (a) $N_m=0$. (b) $N_m=25$. (c) $N_m=49$. (d) $N_m=100$.

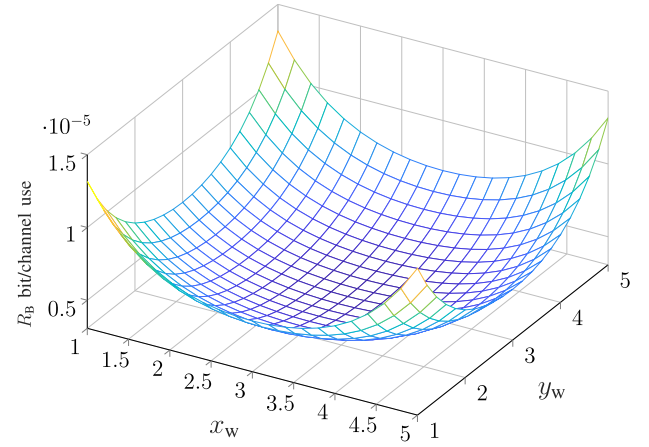


Fig. 8. Case I: Covert transmission rate, R_B , versus Willie's location when $N = 1000$.

(4, 4, 3), and (2,4,3). In this configuration, channel gain superposition from multiple APs at the room center yields higher covert transmission rates in the central region than in other areas.

Furthermore, Fig. 5 and Fig. 6 also demonstrate that the spacing between mRIS elements affects the impact range of the mRIS array on covert transmission rates within the room.

For instance, the element spacing in Fig. 5 (d) is larger by 0.1 m than that in Fig. 6 (d), thus resulting in a wider impact range for the mRIS array in Fig. 5 (d). Moreover, this observation applies to the other subfigures of Fig. 5.

The covert transmission rate at different locations for Case II is illustrated in Fig. 7, in which four APs are uniformly distributed on the ceiling at positions (2, 2, 3), (4, 2, 3), (4, 4, 3), and (2, 4, 3). The conclusions observed in Fig. 5 of Case I also hold for Fig. 7 in Case II. However, due to

the average power constraint in Case II, Fig. 7 demonstrates different covert transmission rate distributions.

Fig. 8 shows the variation of the covert transmission rates with Willie's position in Case I. Covert transmission rate increases significantly when Willie is near room corners due to lower detection channel gains. Notably, when Willie moves to the room center, the superimposition of detection channel gains from four APs causes the covert transmission rate to reach its minimum.

Fig. 9 shows the variation of the covert transmission rates with Willie's position in Case II. Similar to Fig. 8, when Willie is located at room corners, the covert rate significantly increases. Conversely, when Willie is positioned directly below the AP, the elevated detection channel gain reduces the covert rate. Notably, as Willie moves to the room center, although the detection channel gain reaches its maximum, variations in the threshold parameter Ξ_{th} trigger a decrease in $\mathcal{V}_W$. This subsequently induces an enhancement effect through increased A_{II}^* , ultimately leading to an upward trend in covert transmission rates.

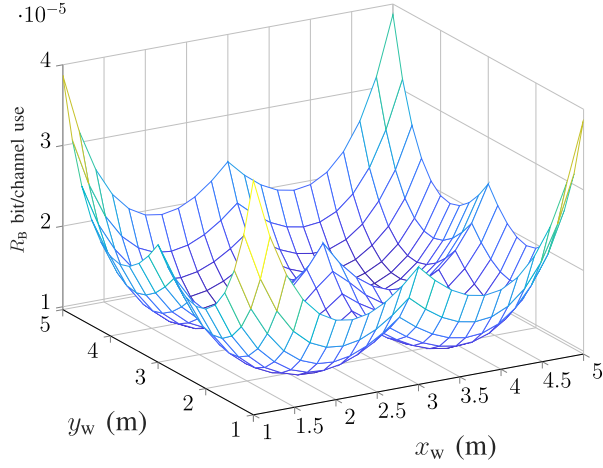


Fig. 9. Case II: Covert transmission rate, R_B , versus Willie's location when $N = 1000$, $\Xi = 0.8$.

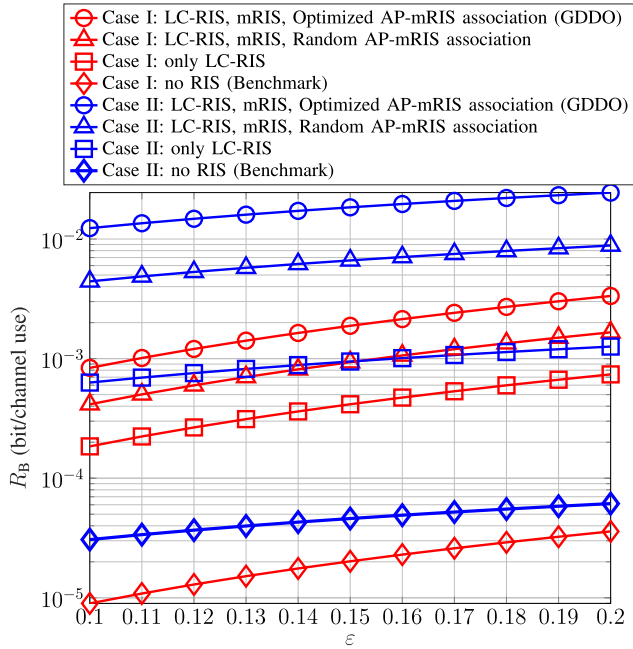


Fig. 10. Covert transmission rate, R_B , versus covertness constraint threshold, ϵ , with different deployments of optical RISs when $N = 1000$, $\lambda = 510$ nm.

Fig. 10 illustrates the relationship between the covert transmission rate R_B of the legitimate user Bob and the covertness constraint threshold ϵ under different deployment schemes of optical RISs. The coordinates of Bob and Willie on the receiving plane are (3, 3, 0.85) and (6, 3, 0.85), respectively. As shown in Fig. 10, the covert transmission rate R_B for all schemes exhibits a declining trend as the covertness constraint ϵ decreases (i.e., under stricter covertness requirements). This trend arises because stricter covertness requirements necessitate tighter constraints on the transmit power to evade detection by Willie. Considering the covert VLC system without any optical RIS as a benchmark, our results demonstrate that the RIS-aided system consistently achieves a higher covert transmission rate. Moreover, the GDDO algorithm achieves a superior covert transmission rate compared to other schemes,

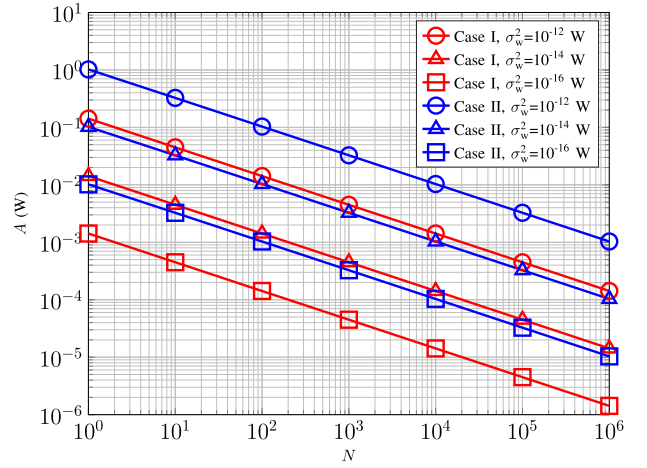


Fig. 11. The maximum peak power, A , versus N , with different case and σ_w .

verifying the effectiveness of the joint optimization, both in case I and case II. Furthermore, Fig. 10 also shows that for the same RIS configurations, the covert transmission rate in Case II consistently exceeds that in Case I. This occurs because the optimal input in Case I follows a uniform distribution, requiring a small power to evade Willie's detection. In contrast, the optimal input in Case II follows a truncated exponential distribution, which inherently concentrates most values near zero, permitting larger power, as shown in Fig. 11.

Fig. 11 shows the peak power, A , versus the number of channel uses N with different noise conditions at Willie. In both Case I and Case II, peak power, A , decreases with increasing N . This occurs because higher N raises Willie's detection probability, necessitating lower transmit power. Besides, lower noise at Willie corresponds to reduced peak power A . This occurs because reduced background noise improves Willie's ability to discriminate the signal from noise, necessitating lower transmit power to blend the signal into the background and avoid detection.

To assess the impact of mRIS and LC-RIS, based on the definition of energy efficiency [26], we define covert energy efficiency as

$$\text{CEE} = \frac{R_{B,n}(A, \eta_c, \gamma, \omega, \mathbf{G}_B)}{P_{\text{total}}}, \quad (97)$$

where P_{total} represents the total system power consumption, given by [26]

$$P_{\text{total}} = P_T + P_{\text{intrinsic}} + P_{\text{mRIS}} + P_{\text{LC}} \quad (98)$$

where $P_{\text{mRIS}} = N_s P_m$ is the power of the mirror array, $P_{\text{intrinsic}}$ denotes the system's intrinsic power, P_T is the transmit power, and P_{LC} represents the LC-RIS power consumption.

The results are shown in Fig. 12, which depicts the CEE versus noise power σ_B^2 under different RIS-assisted schemes. As σ_B^2 increases, the CEE exhibits a downward trend. This is because, although higher noise relaxes the covertness constraint and allows for increased transmit power (see Fig. 11), the deterioration in the overall signal-to-noise ratio (SNR) dominates system performance. Moreover,

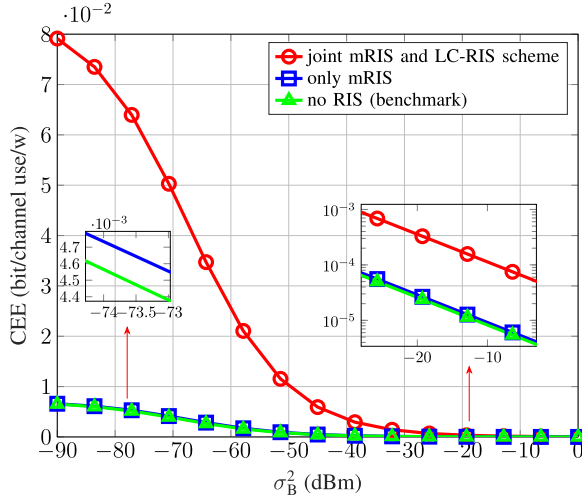


Fig. 12. CEE versus noise power σ_B^2 under different RIS-assisted schemes.

the proposed joint mRIS and LC-RIS scheme consistently achieves a higher CEE than other schemes across the entire noise range. These results validate that integrating LC-RIS effectively enhances covert energy efficiency under varying background noise levels. Furthermore, the mRIS-only scheme shows only marginal CEE improvement over the baseline. This is because the mRIS element, as a passive device, cannot amplify incident signals but only reflects them. Given the low transmit power required by covertness constraints, the signal suffers from severe double-fading attenuation from AP to RIS and RIS to receiver. Consequently, the signal gain at the receiver is limited. Additionally, the power consumption required for mirror rotation further restricts the improvement in CEE.

VI. CONCLUSION

This paper enhances covertness in the MISO-VLC system using dual optical RISs (mRIS/LC-RIS). For the first time, we derive closed-form covertness constraints for the MISO-VLC system in both peak-power-limited scenario and joint peak-and-average-power-limited scenario. Further, we propose a GDDO algorithm to optimize the mRIS and LC-RIS configuration, AP-mRIS association and transmitted power. Simulation results show that the proposed dual optical RIS-assisted MISO-VLC system has a covert rate 10 times higher than that without RISs. In future work, we intend to consider strict per-AP optical power constraints to more closely align with the practical limits of LED driver circuits. Beyond refining the power constraints, we also plan to extend the optical RIS-aided covert VLC framework to more practical settings, particularly scenarios where the eavesdropper is passive and its location or CSI is imperfect or unknown. In this direction, we will investigate robust designs under imperfect or statistical CSI and examine how localization and CSI uncertainty affect covertness and achievable covert rates. We also plan to conduct scalability studies for multi-user scenarios and explore MIMO extensions, aiming to build a more comprehensive and robust covert communication system. In addition, a more general reflected-power model for observation points beyond

the specular-reflection condition will also be considered in our future work.

APPENDIX A

When $\Xi \geq \Xi_{th}$, a peak-power-limited scenario, the input signal $X_W \triangleq \mathbf{h}_W^T \mathbf{x}$ follows a uniform distribution over the interval $(0, \|\mathbf{h}_W^T\|_\infty A]$ [31]. For this distribution, the second moment and the differential entropy are given by

$$\begin{cases} \mathbb{E}[(\mathbf{h}_W^T \mathbf{x})^2] = \frac{1}{3}(\|\mathbf{h}_W^T\|_\infty A)^2, \\ \mathbb{H}(\mathbf{h}_W^T \mathbf{x}) = \log_2(\|\mathbf{h}_W^T\|_\infty A). \end{cases} \quad (99)$$

Substituting Eq. (99) into Eq. (26), we obtain

$$\begin{aligned} \mathcal{Z}_1(A, \mathbf{G}_W) &= \frac{1}{2} \log_2(2\pi\sigma_W^2) + \frac{\frac{1}{3}(\|\mathbf{h}_W^T\|_\infty A)^2 + \sigma_W^2}{2\sigma_W^2 \ln 2} \\ &\quad - \frac{1}{2} \log_2[(\|\mathbf{h}_W^T\|_\infty A)^2 + 2\pi e \sigma_W^2] \leq \frac{2\epsilon^2}{N \ln 2}. \end{aligned} \quad (100)$$

This concludes the proof of the theorem.

APPENDIX B

When $0 < \Xi < \Xi_{th}$, a joint peak-and-average-power-limited scenario, the input signal X_W follows a truncated exponential distribution [31], and its PDF can be expressed as

$$f_{X_W}(X_W) = \frac{p_{W,j}}{h_{W,j} A} \frac{\mu}{1 - e^{-\mu}} e^{-\frac{\mu(X_W - s_{W,j-1} A)}{h_{W,j} A}}, \quad (101)$$

where $X_W \in (s_{W,j-1} A, s_{W,j} A]$, $j \in \{1, 2, \dots, N_s\}$.

Therefore, we have

$$\mathbb{H}(\mathbf{h}_W^T \mathbf{x}) = \mathbb{H}(X_W) = \log(s_{N_s, W} A) + \mathcal{V}_W, \quad (102)$$

where the definition of $\mathcal{V}_W$ is given in Eq. (37).

Meanwhile, $\mathbb{E}((\mathbf{h}_W^T \mathbf{x})^2)$ can be derived as

$$\begin{aligned} &\mathbb{E}((\mathbf{h}_W^T \mathbf{x})^2) \\ &= \mathbb{E}((X_W)^2) \\ &= \sum_{j=1}^{N_s} \int_{s_{W,j-1} A}^{s_{W,j} A} X_W^2 f_{X_W}(X_W) dX_W \\ &= \sum_{j=1}^{N_s} \int_{s_{W,j-1} A}^{s_{W,j} A} X_W^2 \frac{p_{W,j}}{h_{W,j} A} \frac{\mu}{1 - e^{-\mu}} e^{-\frac{\mu(X_W - s_{W,j-1} A)}{h_{W,j} A}} dX_W \\ &= \sum_{j=1}^{N_s} p_{W,j} \frac{\mu A^2}{1 - e^{-\mu}} (t_1 + t_2 + t_3), \end{aligned} \quad (103)$$

$$\begin{aligned} \text{where } t_1 &= s_{W,j-1}^2 \frac{1 - e^{-\mu}}{\mu}, \quad t_2 = 2s_{W,j-1} h_{W,j} \frac{1 - (1 + \mu)e^{-\mu}}{\mu^2}, \\ t_3 &= h_{W,j}^2 \frac{2 - 2(1 + \mu + \frac{\mu^2}{2})e^{-\mu}}{\mu^3}. \end{aligned}$$

Substituting Eqs. (102) and (103) into Eq. (26), we obtain

$$\frac{\sum_{j=1}^{N_s} p_{W,j} \frac{\mu A^2}{1-e^{-\mu}} (t_1 + t_2 + t_3) + \sigma_W^2}{2\sigma_W^2 \ln 2} + \frac{1}{2} \log_2 \frac{2\pi\sigma_W^2}{(s_{W,N_s} A)^2 \cdot 2^{2\nu_W} + 2\pi e \sigma_W^2} \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (104)$$

Upon further organization of the above, we get

$$\begin{aligned} \mathcal{Z}_{\Pi}(A, \mathbf{G}_W) &\triangleq \frac{1}{2\sigma_W^2 \ln 2} \sum_{j=1}^{N_s} p_{W,j} \frac{\mu A^2}{1-e^{-\mu(\lambda)}} (t_1 + t_2 + t_3) \\ &\quad - \frac{1}{2 \ln 2} \ln \left[1 + \frac{(s_{W,N_s} A)^2 \cdot 2^{2\nu_W}}{2\pi e \sigma_W^2} \right] \\ &\leq \frac{2\varepsilon^2}{N \ln 2}. \end{aligned} \quad (105)$$

This concludes the proof of the theorem.

APPENDIX C

From Rotation Angles to Reflective Spot: When the mRIS element k rotates, its normal vector is expressed as

$$\hat{\mathbf{N}}_{j,k}(\gamma_k, \omega_k) = \begin{bmatrix} \sin \omega_k \cos \gamma_k \\ \cos \omega_k \cos \gamma_k \\ \sin \gamma_k \end{bmatrix}. \quad (106)$$

Let $\mathbf{S}_j$ denote the position of AP j , $\mathbf{Q}$ represent an arbitrary point on the receiver plane, and $\mathbf{R}_k$ be the central reflection point of mRIS element k . In accordance with the law of reflection in geometric optics, we can characterize the incident ray direction by

$$\hat{\mathbf{R}}_{j,k}^i = \frac{\mathbf{R}_k - \mathbf{S}_j}{\|\mathbf{R}_k - \mathbf{S}_j\|_2}, \quad (107)$$

and characterize the reflected ray direction by

$$\hat{\mathbf{R}}_{j,k}^r = \frac{\mathbf{Q} - \mathbf{R}_k}{\|\mathbf{Q} - \mathbf{R}_k\|_2}. \quad (108)$$

Then, the reflection law is mathematically formulated as

$$\hat{\mathbf{R}}_{j,k}^r = 2 \cos(\theta_{k,u}) \cdot \hat{\mathbf{N}}_{j,k}(\gamma_k, \omega_k) - \hat{\mathbf{R}}_{j,k}^i, \quad (109)$$

where $\cos(\theta_{k,u}) = \hat{\mathbf{N}}_{j,k}^T(\gamma_k, \omega_k) \cdot \hat{\mathbf{R}}_{j,k}^i$.

Therefore, the rotation angles (γ_k, ω_k) uniquely determine the reflective spot position $\mathbf{Q}$ through (109).

From Reflective Spot to Rotation Angles: Given a reflective spot position $\mathbf{Q}$, the corresponding reflected ray direction is defined as

$$\hat{\mathbf{R}}_{j,k}^r = \frac{\mathbf{Q} - \mathbf{R}_k}{\|\mathbf{Q} - \mathbf{R}_k\|_2}. \quad (110)$$

By combining the incident ray direction $\hat{\mathbf{R}}_{j,k}^i$, the mRIS element's normal vector is derived as

$$\hat{\mathbf{N}}_{j,k}(\mathbf{Q}) = \frac{\hat{\mathbf{R}}_{j,k}^i + \hat{\mathbf{R}}_{j,k}^r}{\|\hat{\mathbf{R}}_{j,k}^i + \hat{\mathbf{R}}_{j,k}^r\|_2}. \quad (111)$$

Subsequently, the rotation angles (γ_k, ω_k) are derived as

$$\begin{cases} \omega_k = \sin^{-1} \left(\hat{\mathbf{N}}_{j,k}(\mathbf{Q}) \cdot \mathbf{e}_3 \right), \\ \gamma_k = \sin^{-1} \left(\frac{\hat{\mathbf{N}}_{j,k}(\mathbf{Q}) \cdot \mathbf{e}_1}{\cos(\omega_k)} \right). \end{cases} \quad (112)$$

Therefore, the position of the reflective spot uniquely maps to the mirror rotation angles.

In conclusion, the position of the reflected light spot and the rotation angles of the mRIS element exhibit a one-to-one mapping.

APPENDIX D

As defined in Section II, the transmission gain of LC-RIS is $\kappa(\eta_c) = \zeta(\eta_c)\ell(\eta_c)$. Under normal incidence, the components are given by

$$\varsigma_0(\eta_c) = \left(\frac{4\eta_a\eta_c}{(\eta_a + \eta_c)^2} \right)^2, \quad (113)$$

and

$$\ell_0(\eta_c) = e^{\frac{2\pi r}{\lambda} \eta_c^3 v(\eta_c)}. \quad (114)$$

To examine concavity, we derive that the derivatives of $\varsigma_0(\eta_c)$ as

$$\frac{d\varsigma_0}{d\eta_c} = \frac{32\eta_a^2\eta_c(\eta_a - \eta_c)}{(\eta_a + \eta_c)^5}, \quad (115)$$

and

$$\frac{d^2\varsigma_0}{d\eta_c^2} = \frac{32\eta_a^2(3\eta_c^2 - 6\eta_a\eta_c + \eta_a^2)}{(\eta_a + \eta_c)^6}. \quad (116)$$

With $\eta_a = 1$, algebraic inspection confirms that $\frac{d\varsigma_0}{d\eta_c} < 0$ and $\frac{d^2\varsigma_0}{d\eta_c^2} < 0$ for $\eta_c \in [1.5, 1.7]$.

Next, the derivatives of $\ell_0(\eta_c)$ are derived as

$$\frac{d\ell_0}{d\eta_c} = \ell_0 \frac{2\pi r}{\lambda} M_1, \quad (117)$$

and

$$\frac{d^2\ell_0}{d\eta_c^2} = \ell_0 \left[\frac{2\pi r}{\lambda} M_1 \right]^2 + \ell_0 \frac{2\pi r}{\lambda} M_2, \quad (118)$$

where $M_1 = 3\eta_c^2 v + \eta_c^3 \frac{dv}{d\eta_c}$ and $M_2 = 6\eta_c v + 6\eta_c^2 \frac{dv}{d\eta_c} + \eta_c^3 \frac{d^2v}{d\eta_c^2}$. Analysis under the simulation parameters in Section V indicates that $\frac{d\ell_0}{d\eta_c}$ exhibits exactly one zero point, and $\frac{d^2\ell_0}{d\eta_c^2} < 0$ for $\eta_c \in [1.5, 1.7]$.

Finally, the second derivative of $\kappa(\eta_c)$ is analyzed using the product rule

$$\frac{d^2\kappa}{d\eta_c^2} = \frac{d^2\varsigma_0}{d\eta_c^2} \ell_0 + \varsigma_0 \frac{d^2\ell_0}{d\eta_c^2} + 2 \frac{d\varsigma_0}{d\eta_c} \frac{d\ell_0}{d\eta_c}. \quad (119)$$

Although the cross-term $\frac{d\varsigma_0}{d\eta_c} \frac{d\ell_0}{d\eta_c}$ is positive, quantitative evaluation demonstrates that it is consistently outweighed by the negative terms within the feasible domain. Thus, $\frac{d^2\kappa}{d\eta_c^2} < 0$ holds, establishing the concavity of $\kappa(\eta_c)$.

APPENDIX E

In the peak-power-limited scenario (Case I), under the null hypothesis $\mathcal{H}_0$, the distribution of the noise, $p_0(y_W)$, is Gaussian and given by

$$p_0(y_W) = \frac{1}{\sqrt{2\pi}\sigma_W} e^{-\frac{y_W^2}{2\sigma_W^2}}. \quad (120)$$

Under hypothesis $\mathcal{H}_1$, based on Eq. (18), the Probability Density Function (PDF) of the signal received at Willie is given by

$$p_1(y_W) = \int_{-\infty}^{\infty} p_s(s) p_{n_W}(y_W - s) ds, \quad (121)$$

where $s \triangleq \mathbf{h}_W^T \mathbf{x}$ denotes the input signal, $p_s(s)$ is the uniform input distribution defined as $p_s(s) = \frac{1}{\|\mathbf{h}_W^T\|_{\infty} A}$ for $0 \leq s \leq \|\mathbf{h}_W^T\|_{\infty} A$ and 0 otherwise, $p_{n_W}(x)$ denotes the noise PDF.

Further, by substituting $p_s(s)$ into Eq. (121), we obtain

$$\begin{aligned} p_1(y_W) &= \int_0^{\|\mathbf{h}_W^T\|_{\infty} A} \frac{1}{\|\mathbf{h}_W^T\|_{\infty} A} \frac{1}{\sqrt{2\pi}\sigma_W} \exp\left(-\frac{(y_W - s)^2}{2\sigma_W^2}\right) ds \\ &= \frac{1}{\|\mathbf{h}_W^T\|_{\infty} A} \frac{1}{\sqrt{2\pi}\sigma_W} \int_0^{\|\mathbf{h}_W^T\|_{\infty} A} \exp\left(-\frac{(y_W - s)^2}{2\sigma_W^2}\right) ds. \end{aligned} \quad (122)$$

Letting $t = (s - y_W) / \sigma_W$ and simplifying, we obtain

$$p_1(y_W) = \frac{1}{\|\mathbf{h}_W^T\|_{\infty} A \sqrt{2\pi}} \int_{-\frac{y_W}{\sigma_W}}^{\frac{\|\mathbf{h}_W^T\|_{\infty} A - y_W}{\sigma_W}} \exp\left(-\frac{t^2}{2}\right) dt. \quad (123)$$

By using the definition of the Gaussian Q-function, we obtain Eq. (96).

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